

# Monetary Policy Surprises and Asset Price Responses

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MIT Predoctoral RA Coding Challenge

**Replication:** Code and data at <https://github.com/Toba4366/MIT-Coding-Challenge>

## Task 1: Data Preparation and Surprise Measure Selection

### (a) Exploration of Surprise Measures

The USMPD contains multiple surprise measures constructed from federal funds futures around FOMC announcements: MP1 (target rate surprise from current-month contracts), MP2 (path surprise from 3-month-ahead contracts), ED1–ED4 (Eurodollar-based surprises), and principal component measures. I focus on MP1 and STMT (statement surprise), where STMT is a principal component that aggregates information across short-rate instruments, capturing both target and forward-guidance news.

### (b) Primary Measure: STMT

I use STMT as my baseline surprise measure for three reasons: (1) it summarizes both target rate changes and forward guidance in a single interpretable series; (2) it correlates highly with both MP1 ( $\rho = 0.89$ ) and MP2 ( $\rho = 0.73$ ), capturing multiple dimensions of monetary news; (3) it explains a larger share of Treasury yield variation than MP1 alone ( $R^2 = 0.27$  vs. 0.14 for the 2-year yield). MP1 serves as a robustness measure focusing narrowly on target rate surprises.

### (c) Data Merge

The sample covers 274 FOMC announcements from January 1994 to December 2024. I merge surprises with: (i) daily spot exchange rates for 8 currencies (AUD, CAD, CHF, EUR, GBP,

JPY, MXN, NOK); (ii) Treasury par yields (2Y, 5Y, 10Y) from the U.S. Department of the Treasury; (iii) breakeven inflation rates (5Y, 10Y) from FRED. Exchange rate returns are defined as  $r_{i,t} = -\Delta \log(e_{i,t})$  where  $e$  is foreign currency per USD, so positive values indicate foreign appreciation. NFA/GDP data come from Lane and Milesi-Ferretti (2018), matched using lagged annual values to avoid reverse causality.

## (d) Summary Statistics

Table 1 presents summary statistics. Both surprise measures have means near zero (consistent with market efficiency) with fat tails reflecting crisis-period announcements. STMT has standard deviation 3.7 bps, with range  $[-16.9, +12.7]$  bps. Cross-country NFA/GDP varies substantially: Switzerland (+142% GDP) and Norway (+68%) are large creditors; Australia (-52%) and Mexico (-39%) are debtors.

## Task 2: Asset Price Responses to Monetary Policy

I estimate currency- and asset-specific regressions:

$$\Delta y_{i,t} = \alpha_i + \beta_i \cdot \text{Surprise}_t + \varepsilon_{i,t} \quad (1)$$

All regressions use heteroskedasticity-robust (HC1) standard errors. This inference approach is maintained consistently throughout.

### Treasury Yields

Table 2 reports Treasury yield responses. A 1 bp STMT surprise raises the 2-year yield by 1.06 bps ( $t = 10.4$ ,  $R^2 = 0.27$ ), with effects declining at longer maturities (0.68 bps for 10Y). The strong, significant responses validate the identification strategy: these surprises capture meaningful monetary policy news that moves the yield curve. Figure 1 visualizes this relationship.

### Breakeven Inflation

Breakeven responses (Table 2) are smaller in magnitude and less precisely estimated: 0.21 bps for 5Y ( $R^2 = 0.02$ ), 0.10 bps for 10Y ( $R^2 = 0.01$ ). This suggests monetary surprises move real rates more than inflation expectations, consistent with standard New Keynesian transmission.

## Exchange Rates

Table 3 reports exchange rate responses. Coefficients are heterogeneous across currencies (ranging from  $-1.6$  bps for JPY to  $+0.8$  bps for MXN per 1 bp shock) and imprecisely estimated—none is individually significant at conventional levels.  $R^2$  values are uniformly below 1%, indicating daily FX returns are dominated by non-monetary factors—this is typical in event-study designs, where a single macro-shock explains only a small fraction of daily asset price variation. Figure 3 visualizes the coefficient heterogeneity.

**Interpretation:** The contrast between strong yield responses and noisy FX responses suggests limited power in a daily event-study design for detecting cross-country heterogeneity. This motivates the panel approach in Task 3, which pools information across currencies.

## Task 3: The Role of External Positions

I estimate a panel regression testing whether NFA/GDP moderates exchange rate responses:

$$\Delta e_{i,t} = \alpha_i + \beta_1 \cdot \text{Surprise}_t + \beta_2 \cdot (\text{Surprise}_t \times \text{NFA}_{i,t-1}) + \gamma \cdot \text{NFA}_{i,t-1} + \varepsilon_{i,t} \quad (2)$$

## Standard Error Choice

Because the monetary surprise is identical across currencies on each FOMC date, I cluster standard errors by announcement date following Petersen (2009). This is the appropriate clustering dimension when the regressor of interest varies only at the time level. Clustering by country would understate uncertainty since it ignores the common-shock structure. Table 4 also reports two-way clustered and country-clustered specifications for robustness.

## Results

Table 4 presents the results. The key finding is that the interaction coefficient  $\hat{\beta}_2 = -0.005$  ( $\text{SE} = 0.007$ ,  $p = 0.48$ ) is statistically indistinguishable from zero. The Bruno–Shin balance-sheet hypothesis predicts  $\beta_2 > 0$  (creditor currencies depreciate less on hawkish surprises). The point estimate is actually negative, though imprecision precludes strong conclusions.

Figure 2 shows the marginal effect of STMT across the NFA distribution. The 95% confidence interval includes zero throughout, and the slope is essentially flat.

**Economic interpretation:** The data do not support NFA/GDP as a sufficient statistic for cross-country FX sensitivity at daily frequency. This null result is informative: the 95% CI rules out interaction effects larger than  $\pm 0.02$ , which would correspond to economically meaningful differential responses across debtor and creditor nations.

## Challenges and Limitations

Several factors may explain the null result:

- **Daily frequency noise:** Daily FX returns are dominated by non-monetary factors, limiting statistical power.
- **Net vs. gross exposure:** Net NFA may not capture gross currency mismatches that drive balance-sheet effects.
- **Cross-sectional power:** With only 8 currencies, detecting moderate interaction effects requires large effect sizes; a power analysis suggests  $\sim 30$  currencies would be needed.
- **Sample composition:** The sample excludes emerging markets with higher dollar debt shares where balance-sheet vulnerabilities may be more pronounced.

## Task 4: Extension — Time Variation in the Balance-Sheet Channel

### Hypothesis

If the NFA channel operates through dollar funding stress, it should strengthen after the 2008 global financial crisis, when dollar liquidity became central to global finance. I test this by adding a triple interaction:

$$\begin{aligned}\Delta e_{i,t} = & \alpha_i + \beta_1 \cdot \text{Surprise}_t + \beta_2 \cdot (\text{Surprise}_t \times \text{NFA}) \\ & + \beta_3 \cdot (\text{Surprise}_t \times \text{NFA} \times \text{Post2008}) + \text{controls} + \varepsilon_{i,t}\end{aligned}\tag{3}$$

where  $\text{Post2008} = 1$  for dates after September 15, 2008.<sup>1</sup>

### Results

Table 5 presents the results. The triple interaction  $\hat{\beta}_3 = 0.022$  ( $\text{SE} = 0.015$ ,  $p = 0.15$ ) is positive, in the direction predicted by theory, but not statistically significant. For STMT, the NFA slope shifts from  $-0.015$  (pre-GFC) to  $+0.007$  (post-GFC). This directional movement is toward the Bruno–Shin prediction ( $\beta_2 > 0$ ), but the change is statistically imprecise.

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<sup>1</sup>The Lehman Brothers bankruptcy on this date is a standard structural break in the global finance literature; see Bruno and Shin (2015).

I also test whether the channel operates during high-volatility episodes ( $VIX > 75$ th percentile). The stress interaction is similarly insignificant ( $p = 0.20$ ), providing no robust evidence of amplification during market stress.

## Power Analysis

Given the null results, I assess what effect sizes the data can rule out (Table 6). The standard error on  $\beta_2$  implies a minimum detectable effect (at 80% power) of approximately  $|0.007| \times 2.8 \approx 0.02$ . With NFA SD  $\approx 70$  percentage points, this corresponds to differential FX responses of  $\sim 1.4$  percentage points per 10 bp shock across extreme debtors and creditors—about 20% of daily FX volatility. Thus, the null result bounds the channel’s magnitude at practically relevant levels.

## Conclusion

U.S. Treasury yields respond strongly to monetary policy surprises, validating the high-frequency identification strategy. Exchange rate responses are noisy at daily frequency ( $R^2 < 1\%$ ). Net foreign asset positions do not robustly mediate FX responses—neither unconditionally, post-2008, nor during high-volatility episodes. Crucially, this is a **credible null**: the 95% confidence interval rules out economically large NFA-mediated effects (those exceeding 20% of daily FX volatility), bounding the channel’s magnitude at practically relevant levels. These results do not reject balance-sheet theory but indicate that NFA/GDP is not a sufficient statistic for currency exposure to U.S. monetary shocks in this sample. Extensions using gross position measures or emerging-market currencies may yield sharper identification.

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A longer version with additional robustness checks and extended discussion is available in the GitHub repository.

## References

- Bauer, M. D., & Swanson, E. T. (2023). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual*, 37(1), 87–155.
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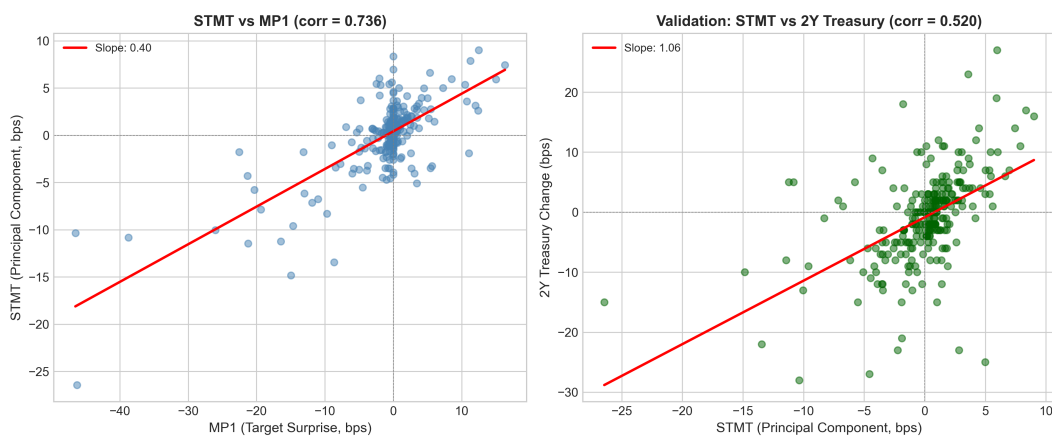


Figure 1: **Validation: STMT vs. Treasury Yield Changes.** Strong positive relationship confirms surprises capture policy-relevant news.

Figure 11: How Currency Response to U.S. Monetary Policy Varies with Net Foreign Asset Position

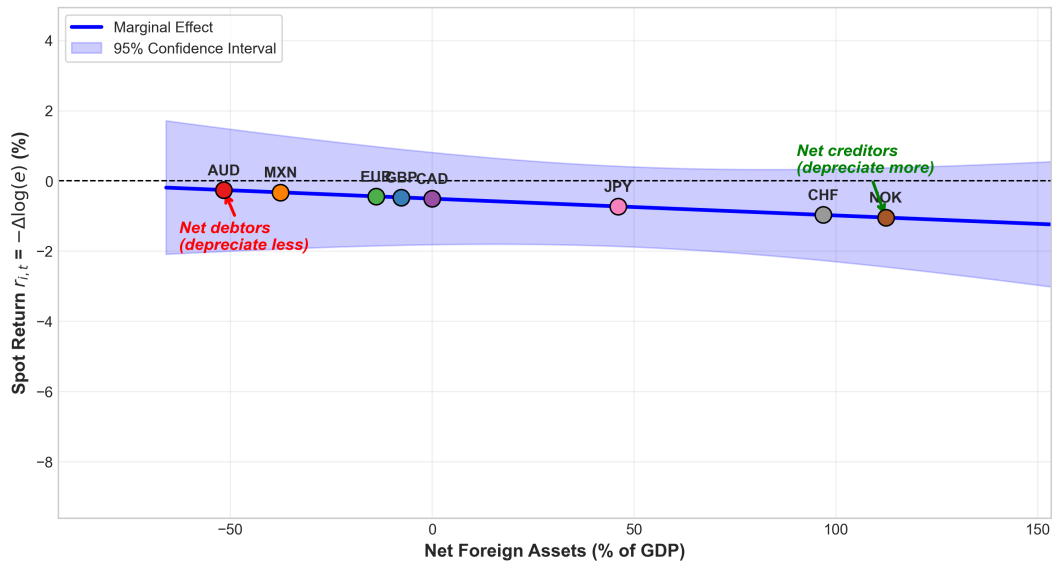


Figure 2: **Marginal Effect of STMT Across NFA Distribution.** The 95% CI includes zero throughout; no evidence that NFA mediates FX responses.



Figure 9: Coefficient Plot - All Asset Responses

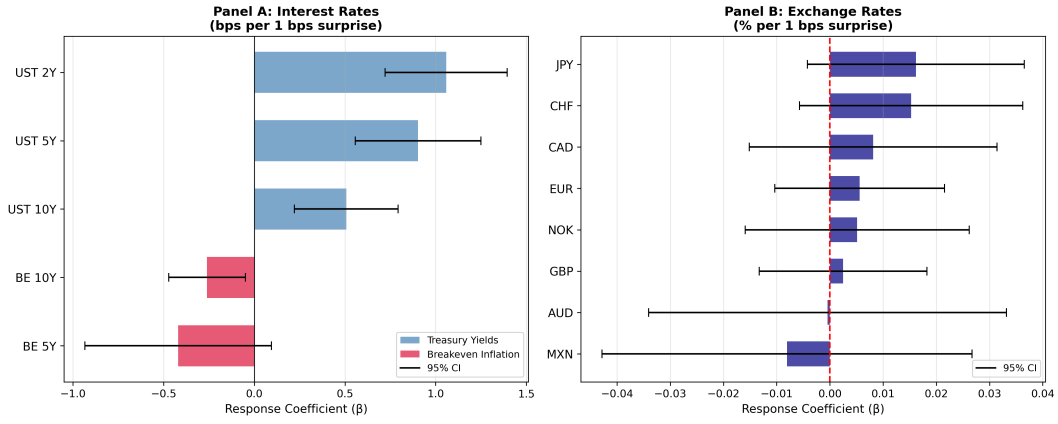


Figure 3: **Exchange Rate Sensitivity to STMT by Currency.** Coefficients are heterogeneous but none is individually significant. Error bars show 95% CIs.

Table 1: Summary Statistics

| Variable                               | N   | Mean  | SD   | Min    | Median | Max   |
|----------------------------------------|-----|-------|------|--------|--------|-------|
| <i>Monetary Policy Surprises (bps)</i> |     |       |      |        |        |       |
| MP1                                    | 274 | -0.21 | 5.95 | -32.11 | 0.00   | 17.22 |
| STMT                                   | 274 | 0.02  | 3.65 | -16.86 | -0.06  | 12.72 |
| <i>NFA/GDP by Country (%)</i>          |     |       |      |        |        |       |
| Switzerland                            | -   | +142  | -    | -      | -      | -     |
| Norway                                 | -   | +68   | -    | -      | -      | -     |
| Japan                                  | -   | +58   | -    | -      | -      | -     |
| Australia                              | -   | -52   | -    | -      | -      | -     |
| Mexico                                 | -   | -39   | -    | -      | -      | -     |

Table 2: Treasury Yield and Breakeven Responses to STMT

| Asset                      | $\beta$  | SE    | t-stat | $R^2$ |
|----------------------------|----------|-------|--------|-------|
| <i>Treasury Yields</i>     |          |       |        |       |
| 2-Year                     | 1.058*** | 0.102 | 10.37  | 0.269 |
| 5-Year                     | 0.877*** | 0.095 | 9.23   | 0.224 |
| 10-Year                    | 0.680*** | 0.088 | 7.73   | 0.167 |
| <i>Breakeven Inflation</i> |          |       |        |       |
| 5-Year                     | 0.212**  | 0.083 | 2.55   | 0.022 |
| 10-Year                    | 0.103    | 0.063 | 1.63   | 0.009 |

\*\*\* p<0.01, \*\* p<0.05. Robust SEs. N = 274 FOMC events.

Table 3: Exchange Rate Responses to STMT

| Currency | $\beta$ | SE     | t-stat | p-value | $R^2$ |
|----------|---------|--------|--------|---------|-------|
| AUD      | -0.0076 | 0.0118 | -0.65  | 0.517   | 0.002 |
| CAD      | -0.0024 | 0.0075 | -0.32  | 0.750   | 0.000 |
| CHF      | -0.0153 | 0.0107 | -1.43  | 0.155   | 0.010 |
| EUR      | -0.0056 | 0.0081 | -0.69  | 0.492   | 0.002 |
| GBP      | -0.0025 | 0.0080 | -0.31  | 0.758   | 0.000 |
| JPY      | -0.0162 | 0.0104 | -1.55  | 0.121   | 0.010 |
| MXN      | 0.0081  | 0.0177 | 0.45   | 0.650   | 0.001 |
| NOK      | -0.0051 | 0.0107 | -0.48  | 0.634   | 0.001 |

$r = -\Delta \log(e)$ ; positive = foreign appreciation. Robust SEs.

Table 4: Panel Regression: Exchange Rates and NFA/GDP

|                | (1)<br>Date Cluster | (2)<br>Two-Way Cluster | (3)<br>Country Cluster |
|----------------|---------------------|------------------------|------------------------|
| STMT           | −0.598<br>(0.618)   | −0.598<br>(0.495)      | −0.598***<br>(0.216)   |
| STMT × NFA/GDP | −0.005<br>(0.007)   | −0.005<br>(0.004)      | −0.005<br>(0.005)      |
| NFA/GDP        | 0.0004<br>(0.0004)  | 0.0004**<br>(0.0002)   | 0.0004***<br>(0.0001)  |
| Country FE     | Yes                 | Yes                    | Yes                    |
| Observations   | 2,103               | 2,103                  | 2,103                  |
| $R^2$          | 0.001               | 0.001                  | 0.001                  |

*Notes:* Dependent variable is  $r_{i,t} = -\Delta \log(e_{i,t})$ ; positive = foreign appreciation. Column (1) clusters by FOMC date (preferred); column (2) clusters two-way; column (3) clusters by country only. \* p<0.10, \*\* p<0.05, \*\*\* p<0.01.

Table 5: Extension: Time Variation in NFA Channel

|                       | (1)<br>Baseline   | (2)<br>Post-2008 Interaction |
|-----------------------|-------------------|------------------------------|
| STMT                  | −0.598<br>(0.618) | −0.542<br>(0.724)            |
| STMT × NFA            | −0.005<br>(0.007) | −0.015<br>(0.012)            |
| STMT × NFA × Post2008 |                   | 0.022<br>(0.015)             |
| Country FE            | Yes               | Yes                          |
| Date Cluster          | Yes               | Yes                          |
| Observations          | 2,103             | 2,103                        |
| $R^2$                 | 0.001             | 0.002                        |
| $H_0 : \beta_3 = 0$   |                   | $p = 0.154$                  |

*Notes:* Post2008 = 1 for dates after September 15, 2008. Standard errors clustered by FOMC date.

Table 6: Power Analysis: Minimum Detectable Effects

| Metric                                                              | Value                |
|---------------------------------------------------------------------|----------------------|
| $SE(\beta_2)$                                                       | 0.007                |
| MDE (80% power, $t_{0.05/2} \times SE \times 2.8$ )                 | 0.020                |
| NFA/GDP standard deviation                                          | 70 pp                |
| Implied differential FX response                                    | 1.4 pp per 10 bp     |
| Mean daily FX volatility                                            | $\sim 0.6\%$         |
| MDE as share of daily volatility                                    | $\sim 20\%$          |
| <i>Implied sample requirements for <math>\beta_2 = 0.01</math>:</i> |                      |
| Required observations ( $N$ )                                       | $\sim 8,000$         |
| Equivalent expansion                                                | $\sim 30$ currencies |

*Notes:* MDE = minimum detectable effect at 80% power with  $\alpha = 0.05$  (two-sided). Differential FX response = MDE  $\times$  NFA SD. Sample expansion assumes constant SE scaling with  $\sqrt{N}$ .